



Dr. ADIL HUSSAIN

PhD in Traffic Information Engineering & Control | Electric Vehicles and Charging Station Optimization, Artificial Intelligence and Intelligent Transportation Systems

Date of birth: 1996/09/21 (29 years, 9 months) | **Nationality:** Pakistani | **Status:** Married

Home Address: House # PD-1415, Dhoke Babu Irfan, Stadium Road, Rawalpindi, Punjab, Pakistan.

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SUMMARY

PhD graduate (Chang'an University, June 2026) specializing in Electric Vehicle (EV) charging station demand analysis and forecasting, with a broader focus on machine learning and deep learning models for detection, classification, analysis, and forecasting tasks across transportation and renewable energy domains.

WORK EXPERIENCE

- Academic Writing Tutor** **2024/09 – 2026/06**

Academic English Writing Workshop by International Education School, Chang'an University

Delivered lectures on research-related topics, assisted other tutors during sessions, guided new students in developing early research skills, and supported students in preparing and publishing research in journals and conferences.

- Android Developer** **2022/03 – 2022/09**

Speckpro Digital, Islamabad, Pakistan

Delivered frontend development and API integration across product- and project-based Android apps, including “Salon Wizz,” the e-commerce app “Kasa Beeston,” and the “CASA Tenant Management System,” after converting a one-week trial into a full-time role.

- **Research Assistant** 2019/09 – 2021/09

Cyber Reconnaissance and Combat Lab (CRC), Islamabad, Pakistan

Conducted research in Computer Science, co-authored and developed research papers, assisted in manuscript submission and revision, and prepared publication-ready documents by updating references and ensuring formatting compliance.

- **Junior Android Developer & Intern** 2019/02 – 2019/09

O3 Interfaces (Pvt) Ltd., Islamabad, Pakistan

Converted a 3-month internship into a full-time role on the “UBL Digital” mobile banking app: analyzed app architecture, implemented and tested REST APIs, performed QA and daily bug fixes, and designed and improved UI/backend features, including bill payments, card management, and digital onboarding (SSDOB).

EDUCATION & TRAINING

- **Ph.D, Traffic Information Engineering and Control** 2022/09 – 2026/06

Chang'an University, Xi'an, China

Supervisor: Professor QingChang Lu

- **Master of Science (MS), Computer Science (MSCS)** 2019/09 – 2022/01

Bahria University, Islamabad, Pakistan

Supervisor: Dr. Kashif Naseer Qureshi

- **Bachelor of Science (BS), Information Technology (BSIT)** 2014/09 – 2018/07

PMAS Arid Agriculture University, Rawalpindi, Pakistan

HONOURS & AWARDS

- **2026 Excellent International Graduate Student** 2026/06

International Education School, Chang'an University, Xi'an, China

- **2025 Star of Study Award** 2025/12

International Education School, Chang'an University, Xi'an, China

- **2025 Excellent International Student** 2025/12

International Education School, Chang'an University, Xi'an, China

- **2024 Outstanding Teaching Assistant** 2025/05

International Education School, Chang'an University, Xi'an, China

- **2024 Star of Study Award** 2024/12

International Education School, Chang'an University, Xi'an, China

SKILLS & EXPERTISE

- **Research:** Scientific writing, literature review, research design, academic publishing, data analysis, dissertation supervision, teaching
- **Technical:** Machine Learning, Deep Learning, Artificial Intelligence
- **Programming:** Python, Java, C#, C++
- **Research Tools:** LaTeX, Origin, QGIS
- **Reference Management:** EndNote, Mendeley, Zotero
- **Tools & IDEs:** Android Studio, Visual Studio, PyCharm, Jupyter, Google Colab
- **Databases:** MS SQL, MySQL, SQLite
- **Productivity:** MS Office (Word, PowerPoint, Excel, Access, OneNote)
- **Versioning & PM:** Bitbucket, GitHub, JIRA, Asana
- **Design & API:** Adobe XD, Zeplin, Canva, Postman, Advanced REST Client
- **Languages:** English (Good) · Chinese (Basic) · Punjabi (Native) · Urdu (Excellent)

JOURNALS, CONFERENCES, SEMINARS & REVIEWER

- **World Transport Convention (WTC):** WTC 2024, Qingdao, China (2024/06) - Poster presentation;
- **Journals served as a Reviewer**
 - **Springer Journals:** Scientific Reports, Discover Artificial Intelligence, Journal of Big Data, International Journal of Data Science and Analytics, International Journal of Machine Learning and Cybernetics, Knowledge and Information Systems.
 - **IEEE Journals:** IEEE Transactions on Smart Grid,
 - **Elsevier Journals:** Electric Power Systems Research, Cluster Computing,
 - **Frontiers Journals:** Frontiers in Artificial Intelligence, Frontiers in Computer Science
 - **Wiley:** Expert Systems
 - **MDPI:** Sustainability
- **Conferences served as the Technical Program Committee (TPC) & Reviewer**
 - 3rd IEEE Beyond Technology Summit on Informatics Int'l Conference (BTS-I2C) 2026
 - 2nd IEEE Beyond Technology Summit on Informatics Int'l Conference (BTS-I2C) 2025
 - 1st IEEE International Conference on Advanced Technologies in Energy and Informatics (ICATEI) 2025
 - 1st IEEE Beyond Technology Summit on Informatics Int'l Conference (BTS-I2C) 2024

PUBLICATIONS

Selected Journal Articles (During Ph.D)

1. **Adil Hussain**, Qing-Chang Lu*, Kashif Naseer Qureshi*, Khalid Javeed. *On-Demand Dynamic Charging Pricing for Electric Vehicles*. [Egyptian Informatics Journal](#) (2026). (Web of Science: SCIE, Q2, IF: 4.2)
2. **Adil Hussain**, Qing-Chang Lu*, Sanam Shahla Rizvi, Shixin Wang, Se Jin Kwon*. *Short Term Demand Forecasting of Electric Vehicle Charging Stations using Context Aware Temporal Transformer Model*. [Scientific Reports](#) (2025) (Web of Science: SCIE, Q1, IF: 4.9)
3. **Adil Hussain***, Vishwanath Eswarakrishnan, Ayesha Aslam, Sajib Tripura. *Charging stations demand forecasting using LSTM based hybrid transformer model*. [Scientific Reports](#) (2025) (Web of Science: SCIE, Q1, IF: 4.9)
4. Abdullahi, Muhammad Rabi, Qing-Chang Lu, **Adil Hussain***, Sajib Tripura, Peng-Cheng Xu, ShiXin Wang. *Location optimization of EV charging stations: A custom K-means cluster algorithm approach*. [Energy Reports](#) (2024) (Web of Science: SCIE, Q2, IF: 6.6)
5. **Adil Hussain**, Qing-Chang Lu*, Kashif Naseer Qureshi. *Context-Aware Temporal Fusion Transformer for Short-Term EV Charging Demand Prediction*. **Sustainable Energy, Grids and Networks** - (Revision Submitted) (Web of Science: SCIE, Q2, IF: 5.8) - Subscription access

Journal Articles – Published

1. Qing-Chang Lu, Shixin Wang, Peng-Cheng Xu, Jing Li, Xu Meng, **Adil Hussain**. *Modeling the Dependency Relationship of Coupled Power and Transportation Network*. [Energy](#) (2025)
2. Ayesha Aslam, Yang Xiaojun, Kashif Naseer Qureshi, **Adil Hussain**, Kayhan Zrar Ghafoor. *GST-Former: GCN-based Spatial Temporal Transformer Model for Interaction-Aware Vehicle Trajectory Prediction*. [Journal of Engineering Research](#) (2025)
3. Faizan Zaman, Zhigang Xu, **Adil Hussain**, Anees Ullah, Khalid Zaman *Enhancing Driver Emotion Recognition Through Deep Ensemble Classification*. [Journal of Intelligent and Connected Vehicles](#) (2025)
4. Sajib Tripura, Qing-Chang Lu, **Adil Hussain**, Tanjim Mehmud. *Optimizing traffic safety message dissemination and resource allocation using adaptive deep reinforcement learning in fog-enabled internet of vehicles network*. [Egyptian Informatics Journal](#) (2025)
5. Sajib Tripura, Qing-Chang Lu, Dhonita Tripura, MD Ibrahim Kholilullah, Arunav Malik Avi, Mostak Ahmed, **Adil Hussain**. *Adaptive Q-Learning-Based Event-Prioritized QoS and Incentive Optimization for Enhancing Safety in Vehicular Fog Networks*. [Egyptian Informatics Journal](#) (2025)
6. Amna Ikram, Sunnia Ikram, El-Sayed M. El-Kenawy, **Adil Hussain**, Amal H. Alharbi, Marwa M. Eid. *A fuzzy-optimized hybrid ensemble model for yield prediction in maize-soybean intercropping system*. [Frontiers in Plant Science](#) (2025)

7. Tariq, Vishwanath Eswarakrishnan, **Adil Hussain**, Zhu Wei, Muhammad Uzair. *[A Novel Energy Replenishment Algorithm to Increase the Network Performance of Rechargeable Wireless Sensor Networks](#)*. **Sensors** (2024)
8. **Adil Hussain***, Ayesha Aslam, Sajib Tripura, Vineet Dhanawat, Varun Shinde. *[Weather Forecasting Using Machine Learning Techniques: Rainfall and Temperature Analysis](#)*. **Journal of Advances in Information Technology** (2024)
9. Kashif Naseer Qureshi, Adi Alhudhaif, **Adil Hussain**, Saleem Iqbal, Gwanggil Jeon. *[Trust aware energy management system for smart homes](#)*. **Computers & Electrical Engineering** (2022)
10. **Adil Hussain**, Kashif Naseer Qureshi, Khalid Javeed, Musaed Alhussein. *[Enhanced Intelligent Intrusion Detection System for E-Commerce](#)*. **Computer Systems Science & Engineering** (2023)
11. Hussain, A., Aslam, A.. *[Hate Speech against Women and Immigrants: A Comparative Analysis of Machine Learning and Text Embedding Techniques](#)*. **Journal of Applied Research and Technology** (2024)
12. **Adil Hussain**, Ayesha Aslam. *[Ensemble-based Approach using Inception V2, VGG-16, and Xception Convolutional Neural Networks for Surface Cracks Detection](#)*. **Journal of Applied Research and Technology** (2024)
13. **Adil Hussain**, Amna Khatoon, Ayesha Aslam, Tariq, Muhammad Asif Khosa. *[A Comparative Performance Analysis of Machine Learning Models for Intrusion Detection Classification](#)*. **Journal of Cyber Security** (2024)
14. Ayesha Aslam, **Adil Hussain**. *[A Performance Analysis of Machine Learning Techniques for Credit Card Fraud Detection](#)*. **Journal of Artificial Intelligence** (2024)
15. **Adil Hussain**, Ayesha Aslam. *[Cardiovascular Disease Prediction Using Risk Factors: A Comparative Performance Analysis of Machine Learning Models](#)*. **Journal of Artificial Intelligence** (2024)
16. **Adil Hussain**, Asad Ullah, Ayesha Aslam, Amna Khatoon. *[A Modified Siamese Network for Facial Assimilation](#)*. **WSEAS Transactions on Signal Processing** (2023)

Journal Articles - Under Review / Revision

1. **Adil Hussain***, Ayesha Aslam, Noman Iqbal, Muhammad Yasir, Sajib Tripura. *Short-Term Charging Station Occupancy Prediction using ToU-based Demand Regime Attention*. **Applied Soft Computing** - (Under Review) (Web of Science: SCIE, Q1, IF: 7.8) - Subscription access
2. **Adil Hussain**, Kashif Naseer Qureshi*, Ayesha Aslam. *EV Charging Demand Prediction using Deep Learning: A Comprehensive Literature Review*. **Journal of Big Data** - (Under Review) (Web of Science: SCIE, Q1, IF: 10.8)
3. Shadab Hanifi, **Adil Hussain***, Asif Khan, Ayesha Aslam. *AI-Based Pavement Distress Assessment: A Critical Review of Sensing, Learning, and Deployment*. **Archives of Computational Methods in Engineering**

4. Yishui Zhu*, Noman Iqbal, **Adil Hussain**, Ying Gao, Luyang Wang. *Reliability-Aware Explainable AI Framework for Multimodal Driver Visual Load Assessment under Dynamic Lighting Conditions*. **IEEE Transactions on Reliability**
5. Noman Iqbal, **Adil Hussain***, Ayesha Aslam, Muhammad Yasir, Lucas Sahani Ludanho. *Short-Term Traffic Congestion Forecasting using Fuzzy C-Means Clustering-based Hybrid Approach*. **Cluster Computing** (Under Review)
6. Sajib Tripura, Qing-Chang Lu, Sanam Shahla Rizvi, **Adil Hussain**, Tae-Sun Chung. *Graph Attention Network-Based Accident Risk Aware Message Dissemination for IoV Safety*. **Expert Systems with Applications** (Revision Submitted)
7. Tariq, Zhu Wei, Kamran Shaukat, Ayesha Aslam, **Adil Hussain**, Md. Mahboob Alam. *A Spatiotemporal Forecast-Guided SAC-Learning Framework for 3D UAV-BS Trajectory Planning*. **Scientific Reports** (Revision Submitted)
8. Ayesha Aslam, Xiaojun Yang, Kashif Naseer Qureshi, Adil Hussain, Nazar Waheed. *CrossTransFuse: Transformer & Cross-Attention Fusion for Lane Change Detection*. **Transportation Research: Interdisciplinary Perspectives** (Revision)

Conference Papers - Published

1. Sajib Tripura, **Adil Hussain**, Mostak Ahamed, Dhonita Tripura, Ayesha Aslam. [*A Dual Attention Mechanism-Based Enhanced Residual Network for Skin Cancer Detection*](#). [IEEE 2nd International Conference on Computing, Applications, and Systems \(COMPAS\)](#) (2025)
2. Ayesha Aslam, Sajib Tripura, Dhonita Tripura, Mostak Ahamed, **Adil Hussain**. [*Households Electricity Load Forecasting using Deep Learning Techniques: An Efficacy Analysis of Attention Mechanism for Hourly Forecasting*](#). [IEEE 2nd International Conference on Computing, Applications, and Systems \(COMPAS\)](#) (2025)
3. **Adil Hussain**, Muhammad Yasir, Noman Iqbal, Sajib Tripura, Ayesha Aslam. [*Heart Disease Classification Using Machine Learning Techniques: An Impact Analysis of GridSearchCV-based Optimization*](#). [IEEE International Conference on Emerging Smart Technologies and Applications \(eSmarTA\)](#) (2025)
4. Ayesha Aslam, **Adil Hussain**, Noman Iqbal, Muhammad Yasir, Sajib Tripura. *Solar Panel Dust Detection: Efficacy Analysis and Optimization using Deep Learning*. [IEEE International Conference on Emerging Smart Technologies and Applications \(eSmarTA\)](#) (2025)
5. **Adil Hussain**, Vineet Dhanawat, Ayesha Aslam, Noman Iqbal, Sajib Tripura. [*Credit Card Fraud Detection Using Machine Learning Techniques: Dealing with Imbalanced Data Using Over-Sampling and Under-Sampling Methods*](#). [IEEE Beyond Technology Summit on Informatics International Conference \(BTS-I2C\)](#) (2024)
6. **Adil Hussain***, Ayesha Aslam. *A Cost-Aware Optimal Charging Time Algorithm for EVs in a Parking Lot*. [World Transport Congress \(WTC\)](#) (2024)

7. **Adil Hussain**, Kashif Naseer Qureshi, Raja Waseem Anwar, Ayesha Aslam. [*A Novel SCD11 CNN Model Performance Evaluation with Inception V3, VGG16 and ResNet50 Using Surface Crack Dataset.*](#) [**IEEE 2nd International Conference on Unmanned Vehicle Systems-Oman \(UVS\)**](#) (2024)
8. **Adil Hussain**, Kashif Naseer Qureshi, Faizan Zaman, Ayesha Aslam. [*Minor Surface Cracks Detection using SCD11 Convolutional Neural Network.*](#) [**IEEE International Conference on Emerging Smart Technologies and Applications \(eSmarTA\)**](#) (2024)
9. **Adil Hussain**, Kashif Naseer Qureshi, Ayesha Aslam, Tariq Tariq, Muhammad Rabiul Abdullahi. [*TSC18 Convolutional Neural Network for Traffic Sign Classification.*](#) [**IEEE International Conference on Emerging Smart Technologies and Applications \(eSmarTA\)**](#) (2024)
10. **Adil Hussain**, Vineet Dhanawat, Ayesha Aslam, Faizan Zaman. [*Electricity Load Forecasting Using Attention-Based Hybrid Deep Learning Model.*](#) [**IEEE International Conference on Artificial Intelligence in Engineering and Technology \(ICAET\)**](#) (2024)
11. Amna Khatoon, Weixing Wang, Asad Ullah, Ishfaq Ahmad, **Adil Hussain**, Mardeni Roslee. [*Advancement in Pavement Condition Assessment: An AI-Based Computer Vision Approach.*](#) [**IEEE Multimedia University Engineering Conference \(MECON\)**](#) (2024)
12. Ayesha Aslam, Sajib Tripura, Muhammad Rabiul Abdullahi, Vineet Dhanawat, **Adil Hussain**. [*Traffic Sign Classification Using German Traffic Sign Dataset Leveraging Vision Transformer \(ViT\) Model.*](#) [**3rd International Conference on Data Science and Emerging Technologies \(DaSet\), Springer**](#) (2024)
13. Faizan Zaman, Zhigang Xu, **Adil Hussain**, Ayesha Aslam, Muhammad Rabiul Abdullahi. [*Concrete and Pavement Cracks Detection Leveraging Vision Transformer \(ViT\) Model.*](#) [**IEEE International Conference on Electrical and Information Technology \(IEIT\)**](#) (2024)
14. **Adil Hussain**, Ayesha Aslam, Tariq. [*Traffic Sign Classification Using a Novel TSC18 Convolutional Neural Network.*](#) [**9th International Conference of Sabaragamuwa University of Sri Lanka.**](#) (2023)
15. Peng-Cheng Xu, Qing-Chang Lu, Feng Tao, Jing Li, **Adil Hussain**, Wang Rundong. *Network-Based Resilience Assessment of Metro Stations: The Relation of Infrastructures and Passengers.* **104th Transportation Research Board (TRB) Annual Meeting** (2025)
16. **Adil Hussain***, Ayesha Aslam. *A Reinforcement Learning based Adaptive Dynamic Pricing for EV Charging Stations (Accepted).* **World Transport Congress (WTC)** (2026)

Conference Papers - Accepted / Under Review

1. Noman Iqbal, Zahirul Islam Kajal, Sajib Tripura, **Adil Hussain**, Ayesha Aslam, Mostak Ahamed. *A Cross-Attention Enhanced BiLSTM-GAT Framework for Spatio-Temporal Traffic Flow Forecasting.* [**IEEE 3rd INTERNATIONAL CONFERENCE ON COMPUTING, APPLICATIONS AND SYSTEMS**](#) (2026)

2. Noman Iqbal, Zahirul Islam Kajal, Sajib Tripura, **Adil Hussain**, Ayesha Aslam, Mostak Ahamed. *Traffic Speed Prediction using LSTM and Spatial GCN-based Feature Fusion*. [IEEE 3rd INTERNATIONAL CONFERENCE ON COMPUTING, APPLICATIONS AND SYSTEMS](#) (2026)

Book Chapters

1. **Adil Hussain**, Ayesha Aslam, Rundong Wang, Kashif Naseer Qureshi. *Advanced Learning Techniques for AI in Renewable Energy* (Chapter 7). [Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks](#) 1st Edition, Elsevier, eBook ISBN: 9780443406195.
2. Ayesha Aslam, **Adil Hussain**, Kashif Naseer Qureshi. *Big Data Analytics for AI in Renewable Energy Networks* (Chapter 9). [Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks](#) 1st Edition, Elsevier, eBook ISBN: 9780443406195.
3. **Adil Hussain**, Kashif Naseer Qureshi, Ayesha Aslam. *Novel Machine and Deep Learning and Training Techniques for AI-based Healthcare Systems* (Chapter 7). [Artificial Intelligence-based Smart Healthcare Systems: New Standards, Technologies and Communication Systems](#) 1st Edition, Elsevier, Paperback ISBN: 9780443264764.
4. Ayesha Aslam, Kashif Naseer Qureshi, **Adil Hussain**. *Advanced AI-Based Healthcare Systems Applications and Services* (Chapter 4). [Next Generation AI Language Models in Research: Promising Perspectives and Valid Concerns](#) 1st Edition. Taylor & Francis - CRC Press, eBook ISBN: 9781032667911.
5. **Adil Hussain**, Peng-Cheng Xu, Wang Shixin, Kashif Naseer Qureshi. *Advanced AI-Based Healthcare Systems Applications and Services* (Chapter 6). [Next Generation AI Language Models in Research: Promising Perspectives and Valid Concerns](#) 1st Edition. Taylor & Francis - CRC Press, eBook ISBN: 9781032667911.
6. **Adil Hussain**, Kashif Naseer Qureshi. *Standards and Policies Adoption for AIoT Networks* (Chapter 4). [Artificial Intelligence of Things \(AIoT\): New Standards, Technologies and Communication Systems](#), 1st Edition, CRC Press, eBook ISBN: 9781003430018.
7. **Adil Hussain**, Ayesha Aslam, Mahad Ahmad. *AI in Agriculture - Applications, Challenges, and Future Directions* (Chapter 2, In Press). **AI-Driven Farming Systems With Minimal Human Intervention**, 1st Edition: IGI-Global, eISBN13: 9798337392974.

REFERENCES

- The reference details are provided on the following page.
- Reference emails can be furnished upon request.

LIST OF REFEREES

Ph.D Supervisor:

Professor Qing-Chang Lu

Chang'an Scholar Chair Professor, Vice Dean / School of Electronics and Control Engineering
Department Chair / Department of Traffic Information and Control Engineering, Chang'an University.

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***New Affiliation:** School of Management, Shanghai University, Shanghai, China.

Email: qclu@shu.edu.cn

Master's Supervisor:

Dr. Kashif Naseer Qureshi

Associate Professor, Department of Computer Science, Bahria University, Islamabad, Pakistan.

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